

Sri Lanka Institute of Information Technology

B.Sc. (Hons) Information Technology



Specialization in Information Technology

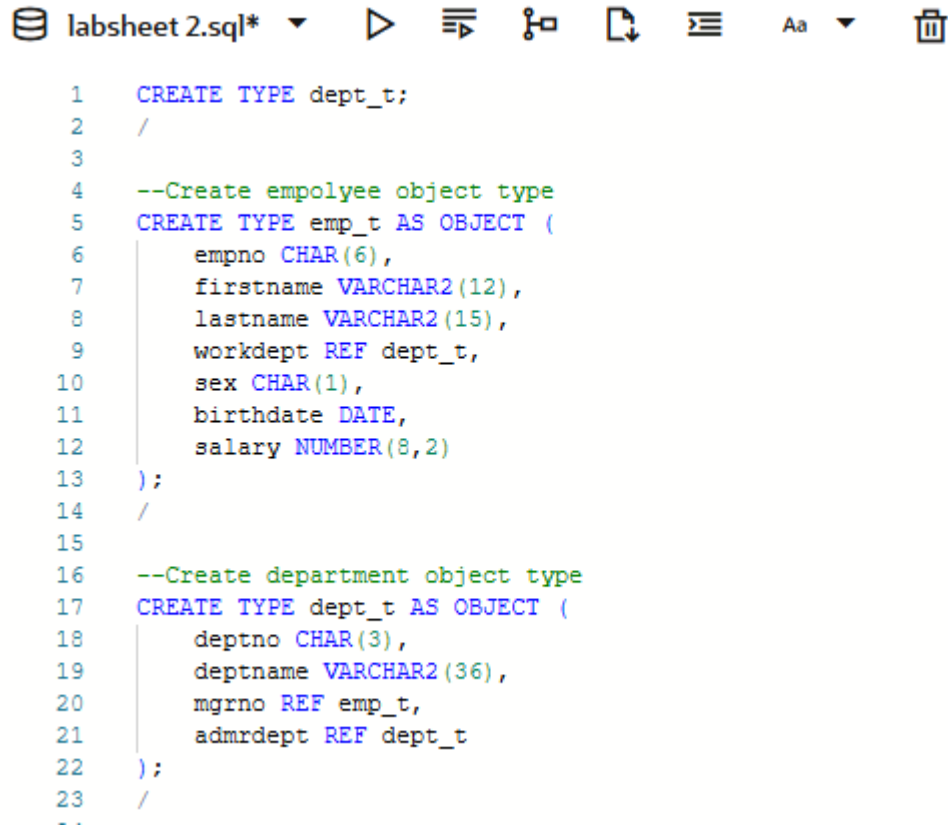
IT3020 - Database Systems

Laboratory Worksheet 02

Student ID: IT23421226

Question 01

(a) Define object types emp_t and dept_t with attributes of EMP and DEPT respectively. Use REF types for workdept (of EMP), and mgrno and admrdept (of DEPT).



```

1  CREATE TYPE dept_t;
2  /
3
4  --Create empolyee object type
5  CREATE TYPE emp_t AS OBJECT (
6      empno CHAR(6),
7      firstname VARCHAR2(12),
8      lastname VARCHAR2(15),
9      workdept REF dept_t,
10     sex CHAR(1),
11     birthdate DATE,
12     salary NUMBER(8,2)
13 );
14 /
15
16 --Create department object type
17 CREATE TYPE dept_t AS OBJECT (
18     deptno CHAR(3),
19     deptname VARCHAR2(36),
20     mgrno REF emp_t,
21     admrdept REF dept_t
22 );
23 /
  
```

Output:

Statement

Executed on: Feb 8, 2026, 8:38:52 AM

```

CREATE TYPE emp_t AS OBJECT (
  empno CHAR(6),
  firstname VARCHAR2(12),
  lastname VARCHAR2(15),
  workdept REF dept_t,
  sex CHAR(1),
  birthdate DATE,
  salary NUMBER(8,2)
);
  
```

 Insert Into Editor

Result

Type EMP_T compiled
Elapsed: 00:00:00.054

Statement

Executed on: Feb 8, 2026, 8:29:22 AM

```

CREATE TYPE dept_t AS OBJECT (
  deptno CHAR(3),
  deptname VARCHAR2(36),
  mgrno REF emp_t,
  admrdept REF dept_t
);
  
```

 Insert Into Editor

Result

Type DEPT_T compiled

Errors for TYPE DEPT_T:

LINE/COL ERROR

```

0/0      PL/SQL: Compilation unit analysis terminated
4/15     PLS-00201: identifier 'EMP_T' must be declared
Elapsed: 00:00:00.055
  
```

(b) Create tables (named as OREMP and ORDEPT) using the object types defined in (a), with appropriate primary keys and referential constraints.

Create department and employee object table

```
25  --create department object table
26  CREATE TABLE ORDEPT OF dept_t (
27  |    deptno PRIMARY KEY
28  );
29
30  --create employee table
31  CREATE TABLE OREMP OF emp_t (
32  |    empno PRIMARY KEY
33  );
34
```

Output:

Statement

Executed on: Feb 8, 2026, 9:26:58 AM

```
CREATE TABLE OREMP OF emp_t (
  empno PRIMARY KEY
)
```



 Insert Into Editor

Result

Table OREMP created.
Elapsed: 00:00:00.023



Statement

Executed on: Feb 8, 2026, 9:25:43 AM

```
CREATE TABLE ORDEPT OF dept_t (
  deptno PRIMARY KEY
)
```



 Insert Into Editor

Result

Table ORDEPT created.
Elapsed: 00:00:00.029



(c) Insert data into the object relational tables created in (b), using the data in EMP and DEPT tables given below. (First insert rows into oremp with null for workdept and rows into ordept with null for admrdept. Then update oremp and ordept with actual REF values for workdept and admrdept.)

Insert rows into OREMP with null

```

35  --insert employee
36  INSERT INTO OREMP VALUES
37  ('000010','CHRISTINE','HAAS',NULL,'F',
38   DATE '1953-08-14',72750);
39
40  INSERT INTO OREMP VALUES
41  ('000020','MICHAEL','THOMPSON',NULL,'M',
42   DATE '1968-02-02',61250);
43
44  INSERT INTO OREMP VALUES
45  ('000030','SALLY','KWAN',NULL,'F',
46   DATE '1971-05-11',58250);
47
48  INSERT INTO OREMP VALUES
49  ('000060','IRVING','STERN',NULL,'M',
50   DATE '1965-07-07',55555);
51
52  INSERT INTO OREMP VALUES
53  ('000070','EVA','PULASKI',NULL,'F',
54   DATE '1973-05-26',56170);
55
56  INSERT INTO OREMP VALUES
57  ('000050','JOHN','GEYER',NULL,'M',
58   DATE '1955-09-15',60175);
59
60  INSERT INTO OREMP VALUES
61  ('000090','EILEEN','HENDERSON',NULL,'F',
62   DATE '1961-05-15',49750);
63
64  INSERT INTO OREMP VALUES
65  ('000100','THEODORE','SPENSER',NULL,'M',
66   DATE '1976-12-18',46150);

```

Insert rows into ORDEPT with null

```

68  --insert department
69  INSERT INTO ORDEPT VALUES
70  ('A00','SPIFFY COMPUTER SERVICE DIV.',NULL,NULL);
71
72  INSERT INTO ORDEPT VALUES
73  ('B01','PLANNING',NULL,NULL);
74
75  INSERT INTO ORDEPT VALUES
76  ('C01','INFORMATION CENTER',NULL,NULL);
77
78  INSERT INTO ORDEPT VALUES
79  ('D01','DEVELOPMENT',NULL,NULL);
80

```

Output:

Statement

Executed on: Feb 8, 2026, 9:50:17 AM

```
INSERT INTO ORDEPT VALUES
('D01','DEVELOPMENT',NULL,NULL)
```



 Insert Into Editor

Result

1 row inserted.

Elapsed: 00:00:00.002



Update OREMP and ORDEPT with actual REF values for workdept and admrdept.

```
81  --update employee work department
82  UPDATE OREMP e
83  SET workdept = (
84    SELECT REF(d)
85    FROM ORDEPT d
86    WHERE d.deptno='A00'
87  )
88  WHERE e.empno='000010';
89
90  UPDATE OREMP e
91  SET workdept = (SELECT REF(d) FROM ORDEPT d WHERE d.deptno='B01')
92  WHERE e.empno IN ('000020','000090','000100');
93
94  UPDATE OREMP e
95  SET workdept = (SELECT REF(d) FROM ORDEPT d WHERE d.deptno='C01')
96  WHERE e.empno IN ('000030','000050');
97
98  UPDATE OREMP e
99  SET workdept = (SELECT REF(d) FROM ORDEPT d WHERE d.deptno='D01')
100 WHERE e.empno IN ('000060','000070');
101
102  --update department managers
103  UPDATE ORDEPT d
104  SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000010')
105  WHERE d.deptno='A00';
106
107  UPDATE ORDEPT d
108  SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000020')
109  WHERE d.deptno='B01';
110
111  UPDATE ORDEPT d
112  SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000030')
113  WHERE d.deptno='C01';
114
115  UPDATE ORDEPT d
116  SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000060')
117  WHERE d.deptno='D01';
118
119  --
```

```

101
102 --update department managers
103 UPDATE ORDEPT d
104 SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000010')
105 WHERE d.deptno='A00';
106
107 UPDATE ORDEPT d
108 SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000020')
109 WHERE d.deptno='B01';
110
111 UPDATE ORDEPT d
112 SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000030')
113 WHERE d.deptno='C01';
114
115 UPDATE ORDEPT d
116 SET mgrno = (SELECT REF(e) FROM OREMP e WHERE e.empno='000060')
117 WHERE d.deptno='D01';
118
119 --update administrative department
120 UPDATE ORDEPT d
121 SET admrdept = (SELECT REF(x) FROM ORDEPT x WHERE x.deptno='A00')
122 WHERE d.deptno='B01';
123
124 UPDATE ORDEPT d
125 SET admrdept = (SELECT REF(x) FROM ORDEPT x WHERE x.deptno='B01')
126 WHERE d.deptno='C01';
127
128 UPDATE ORDEPT d
129 SET admrdept = (SELECT REF(x) FROM ORDEPT x WHERE x.deptno='C01')
130 WHERE d.deptno='D01';
131
132

```

Output:

Statement

Executed on: Feb 8, 2026, 10:02:01 AM

```

UPDATE ORDEPT d
SET admrdept = (SELECT REF(x) FROM ORDEPT x WHERE x.deptno='C01')
WHERE d.deptno='D01'

```



Insert Into Editor

Result

1 row updated.

Elapsed: 00:00:00.004



Question 02

(a) Get the department name and manager's lastname for all departments

Get department name and manager's last name for all departments

```

123
124  --Get department name and manager's last name for all departments
125  SELECT d.deptname,
126         Deref(d.mgrno).lastname AS manager_lastname
127  FROM ORDEPT d;
128

```

Output:

Download ▾ Execution time: 0.012 seconds		
	DEPTNAME	MANAGER_LASTNA
1	SPIFFY COMPUTER S	HAAS
2	PLANNING	THOMPSON
3	INFORMATION CENT	KWAN
4	DEVELOPMENT	STERN

(b) Get the employee number, last name and the department name of every employee

```

128
129  --Get employee number, lastname, and department name of every employee.
130  SELECT e.empno,
131         e.lastname,
132         Deref(e.workdept).deptname AS department_name
133  FROM OREMP e;
134

```

Download ▾ Execution time: 0.009 seconds

Output:

	EMPNO	LASTNAME	DEPARTMENT_NAME
1	000010	HAAS	SPIFFY COMPUTER SERVICE DIV.
2	000020	THOMPSON	PLANNING
3	000030	KWAN	INFORMATION CENTER
4	000060	STERN	DEVELOPMENT
5	000070	PULASKI	DEVELOPMENT
6	000050	GEYER	INFORMATION CENTER
7	000090	HENDERSON	PLANNING
8	000100	SPENSER	PLANNING

(c) For each department, display the department number, department name, and name of the administrative department.

```

135
136  SELECT d.deptno,
137         d.deptname,
138         Deref(d.admrdept).deptname AS admin_department
139  FROM ORDEPT d;
140

```

Output:

  Download Execution time: 0.008 seconds

	DEPTNO	DEPTNAME	ADMIN_DEPARTMENT
1	A00	SPIFFY COMPUTER SERVICE DIV.	(null)
2	B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.
3	C01	INFORMATION CENTER	PLANNING
4	D01	DEVELOPMENT	INFORMATION CENTER

(d) For each department, display the department number, department name, the name of the administrative department and the last name of the manager of the administrative department

```

140
141  /*display the department number, department name, the name of the administrative department
142  and the last name of the manager of the administrative department*/
143  SELECT d.deptno,
144         d.deptname,
145         Deref(d.admrdept).deptname AS admin_department,
146         Deref(Deref(d.admrdept).mgrno).lastname AS admin_manager_lastname
147  FROM ORDEPT d;
148

```

Output:

  Download Execution time: 0.007 seconds

	DEPTNO	DEPTNAME	ADMIN_DEPARTMENT	ADMIN_MANAGER_LASTNAME
1	A00	SPIFFY COMPUTER SERVICE DIV.	(null)	(null)
2	B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.	HAAS
3	C01	INFORMATION CENTER	PLANNING	THOMPSON
4	D01	DEVELOPMENT	INFORMATION CENTER	KWAN

(e) Display the employee number, firstname, lastname and salary of every employee, along with lastname and salary of the manager of the employee's work department.

```

149  /*Employee number, first name, last name, salary
150  Last name and salary of the manager of the employee's work department*/
151  SELECT e.empno,
152         e.firstname,
153         e.lastname,
154         e.salary AS emp_salary,
155         Deref(Deref(e.workdept).mgrno).lastname AS manager_lastname,
156         Deref(Deref(e.workdept).mgrno).salary AS manager_salary
157  FROM OREMP e;
158

```

Output:

  Download ▾ Execution time: 0.007 seconds

	EMPNO	FIRSTNAME	LASTNAME	EMP_SALARY	MANAGER_LASTNAME	MANAGER_SALARY
1	000010	CHRISTINE	HAAS	72750	HAAS	72750
2	000020	MICHAEL	THOMPSON	61250	THOMPSON	61250
3	000030	SALLY	KWAN	58250	KWAN	58250
4	000060	IRVING	STERN	55555	STERN	55555
5	000070	EVA	PULASKI	56170	STERN	55555
6	000050	JOHN	GEYER	60175	KWAN	58250
7	000090	EILEEN	HENDERSON	49750	THOMPSON	61250
8	000100	THEODORE	SPENSER	46150	THOMPSON	61250

(f) Show the average salary for men and the average salary for women for each department. Identify the department by both department number and name.

```

158
159  /*Show the average salary for men and the average salary for women for each department. Identify the
160  department by both department number and name.*/
161  SELECT Deref(e.workdept).deptno AS dept_no,
162         Deref(e.workdept).deptname AS dept_name,
163         AVG(CASE WHEN e.sex = 'M' THEN e.salary END) AS avg_male_salary,
164         AVG(CASE WHEN e.sex = 'F' THEN e.salary END) AS avg_female_salary
165  FROM OREMP e
166  GROUP BY Deref(e.workdept).deptno,
167         Deref(e.workdept).deptname;
168

```

Output:

  Download ▾ Execution time: 0.014 seconds

	DEPT_NO	DEPT_NAME	AVG_MALE_SALARY	AVG_FEMALE_SALARY
1	A00	SPIFFY COMPUTER SERVICE DIV.	(null)	72750
2	B01	PLANNING	53700	49750
3	C01	INFORMATION CENTER	60175	58250
4	D01	DEVELOPMENT	55555	56170

Question 03

```
SQL> @C:\Users\User\Downloads\script
```

DEPTNAME	LASTNAME
SPIFFY COMPUTER SERVICE DIV.	HAAS
PLANNING	THOMPSON
INFORMATION CENTRE	KWAN
DEVELOPMENT CENTRE	STERN

EMPNO	LASTNAME	DEPTNAME
000010	HAAS	SPIFFY COMPUTER SERVICE DIV.
000020	THOMPSON	PLANNING
000030	KWAN	INFORMATION CENTRE
000060	STERN	DEVELOPMENT CENTRE
000070	PULASKI	DEVELOPMENT CENTRE
000050	GEYER	INFORMATION CENTRE
000090	HENDERSON	PLANNING
000100	SPENSER	PLANNING

```
8 rows selected.
```

DEP	DEPTNAME	DEPTNAME
A00	SPIFFY COMPUTER SERVICE DIV.	SPIFFY COMPUTER SERVICE DIV.
B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.
C01	INFORMATION CENTRE	SPIFFY COMPUTER SERVICE DIV.
D01	DEVELOPMENT CENTRE	INFORMATION CENTRE

DEP	DEPTNAME	DEPTNAME
A00	SPIFFY COMPUTER SERVICE DIV. HAAS	SPIFFY COMPUTER SERVICE DIV.
B01	PLANNING HAAS	SPIFFY COMPUTER SERVICE DIV.
C01	INFORMATION CENTRE HAAS	SPIFFY COMPUTER SERVICE DIV.

DEP	DEPTNAME	DEPTNAME
D01	DEVELOPMENT CENTRE KWAN	INFORMATION CENTRE

EMPNO	FIRSTNAME	LASTNAME	SALARY	LASTNAME	SALARY
000010	CHRISTINE	HAAS	72750	HAAS	72750
000020	MICHAEL	THOMPSON	61250	THOMPSON	61250
000030	SALLY	KWAN	58250	KWAN	58250
000060	IRVING	STERN	55555	STERN	55555
000070	EVA	PULASKI	56170	STERN	55555
000050	JOHN	GEYER	60175	KWAN	58250
000090	EILEEN	HENDERSON	49750	THOMPSON	61250
000100	THEODORE	SPENSER	46150	THOMPSON	61250

8 rows selected.

DEP	DEPTNAME	S	AVG(E.SALARY)
A00	SPIFFY COMPUTER SERVICE DIV.	F	72750
B01	PLANNING	M	53700
C01	INFORMATION CENTRE	F	58250
D01	DEVELOPMENT CENTRE	M	55555
D01	DEVELOPMENT CENTRE	F	56170
C01	INFORMATION CENTRE	M	60175
B01	PLANNING	F	49750

7 rows selected.

```
SQL> set echo on
SQL> @C:\Users\User\Downloads\script
SQL> -- IT3020 Practical 02
SQL> -- Exercise 2 : Query Script
SQL>
SQL> -- (a) Department name and manager last name
SQL> SELECT d.deptname, m.lastname
2 FROM ORDEPT d, OREMP m
3 WHERE d.mgrno = REF(m)
4 /
```

DEPTNAME	LASTNAME
SPIFFY COMPUTER SERVICE DIV.	HAAS
PLANNING	THOMPSON
INFORMATION CENTRE	KWAN
DEVELOPMENT CENTRE	STERN

```
SQL>
SQL> -- (b) Employee number, last name, and department name
SQL> SELECT e.empno, e.lastname, d.deptname
2 FROM OREMP e, ORDEPT d
3 WHERE e.workdept = REF(d)
4 /
```

EMPNO	LASTNAME	DEPTNAME
000010	HAAS	SPIFFY COMPUTER SERVICE DIV.
000020	THOMPSON	PLANNING
000030	KWAN	INFORMATION CENTRE
000060	STERN	DEVELOPMENT CENTRE
000070	PULASKI	DEVELOPMENT CENTRE
000050	GEYER	INFORMATION CENTRE
000090	HENDERSON	PLANNING
000100	SPENSER	PLANNING

8 rows selected.

```
SQL>
SQL> -- (c) Department number, department name, and administrative department name
SQL> SELECT d.deptno, d.deptname, a.deptname
  2 FROM ORDEPT d, ORDEPT a
  3 WHERE d.admrdept = REF(a)
  4 /
```

DEP	DEPTNAME	DEPTNAME
A00	SPIFFY COMPUTER SERVICE DIV.	SPIFFY COMPUTER SERVICE DIV.
B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.
C01	INFORMATION CENTRE	SPIFFY COMPUTER SERVICE DIV.
D01	DEVELOPMENT CENTRE	INFORMATION CENTRE

```
SQL>
SQL> -- (d) Department details with admin department manager
SQL> SELECT d.deptno, d.deptname, a.deptname, m.lastname
  2 FROM ORDEPT d, ORDEPT a, OREMP m
  3 WHERE d.admrdept = REF(a)
  4 AND a.mgrno = REF(m)
  5 /
```

DEP	DEPTNAME	DEPTNAME	LASTNAME
A00	SPIFFY COMPUTER SERVICE DIV.	SPIFFY COMPUTER SERVICE DIV.	HAAS
B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.	HAAS
C01	INFORMATION CENTRE	SPIFFY COMPUTER SERVICE DIV.	HAAS

DEP	DEPTNAME	DEPTNAME	LASTNAME
D01	DEVELOPMENT CENTRE	INFORMATION CENTRE	KWAN

```
SQL>
SQL> -- (e) Employee salary with manager salary
SQL> SELECT e.empno, e.firstname, e.lastname, e.salary,
  2 m.lastname, m.salary
  3 FROM OREMP e, ORDEPT d, OREMP m
  4 WHERE e.workdept = REF(d)
  5 AND d.mgrno = REF(m)
  6 /
```

EMPNO	FIRSTNAME	LASTNAME	SALARY	LASTNAME	SALARY
000010	CHRISTINE	HAAS	72750	HAAS	72750
000020	MICHAEL	THOMPSON	61250	THOMPSON	61250
000030	SALLY	KWAN	58250	KWAN	58250
000060	IRVING	STERN	55555	STERN	55555
000070	EVA	PULASKI	56170	STERN	55555
000050	JOHN	GEYER	60175	KWAN	58250
000090	EILEEN	HENDERSON	49750	THOMPSON	61250
000100	THEODORE	SPENSER	46150	THOMPSON	61250

8 rows selected.

```

SQL>
SQL> -- (f) Average salary by sex per department
SQL> SELECT d.deptno, d.deptname, e.sex, AVG(e.salary)
      2 FROM OREMP e, ORDEPT d
      3 WHERE e.workdept = REF(d)
      4 GROUP BY d.deptno, d.deptname, e.sex
      5 /

```

DEP	DEPTNAME	S	AVG(E.SALARY)
A00	SPIFFY COMPUTER SERVICE DIV.	F	72750
B01	PLANNING	M	53700
C01	INFORMATION CENTRE	F	58250
D01	DEVELOPMENT CENTRE	M	55555
D01	DEVELOPMENT CENTRE	F	56170
C01	INFORMATION CENTRE	M	60175
B01	PLANNING	F	49750

7 rows selected.

```

SQL> set termout on
SQL> @C:\Users\User\Downloads\script
SQL> -- IT3020 Practical 02
SQL> -- Exercise 2 : Query Script
SQL>
SQL> -- (a) Department name and manager last name
SQL> SELECT d.deptname, m.lastname
      2 FROM ORDEPT d, OREMP m
      3 WHERE d.mgrno = REF(m)
      4 /

```

DEPTNAME	LASTNAME
SPIFFY COMPUTER SERVICE DIV.	HAAS
PLANNING	THOMPSON
INFORMATION CENTRE	KWAN
DEVELOPMENT CENTRE	STERN

```
SQL>
SQL> -- (b) Employee number, last name, and department name
SQL> SELECT e.empno, e.lastname, d.deptname
2 FROM OREMP e, ORDEPT d
3 WHERE e.workdept = REF(d)
4 /
```

EMPNO	LASTNAME	DEPTNAME
000010	HAAS	SPIFFY COMPUTER SERVICE DIV.
000020	THOMPSON	PLANNING
000030	KWAN	INFORMATION CENTRE
000060	STERN	DEVELOPMENT CENTRE
000070	PULASKI	DEVELOPMENT CENTRE
000050	GEYER	INFORMATION CENTRE
000090	HENDERSON	PLANNING
000100	SPENSER	PLANNING

8 rows selected.

```
SQL>
SQL> -- (c) Department number, department name, and administrative department name
SQL> SELECT d.deptno, d.deptname, a.deptname
2 FROM ORDEPT d, ORDEPT a
3 WHERE d.admrdept = REF(a)
4 /
```

DEP	DEPTNAME	DEPTNAME
A00	SPIFFY COMPUTER SERVICE DIV.	SPIFFY COMPUTER SERVICE DIV.
B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.
C01	INFORMATION CENTRE	SPIFFY COMPUTER SERVICE DIV.
D01	DEVELOPMENT CENTRE	INFORMATION CENTRE

```
SQL>
SQL> -- (d) Department details with admin department manager
SQL> SELECT d.deptno, d.deptname, a.deptname, m.lastname
2 FROM ORDEPT d, ORDEPT a, OREMP m
3 WHERE d.admrdept = REF(a)
4 AND a.mgrno = REF(m)
5 /
```

DEP	DEPTNAME	DEPTNAME	LASTNAME
A00	SPIFFY COMPUTER SERVICE DIV.	SPIFFY COMPUTER SERVICE DIV.	HAAS
B01	PLANNING	SPIFFY COMPUTER SERVICE DIV.	HAAS
C01	INFORMATION CENTRE	SPIFFY COMPUTER SERVICE DIV.	HAAS

```

DEP DEPTNAME                                DEPTNAME
---
LASTNAME
-----
D01 DEVELOPMENT CENTRE                      INFORMATION CENTRE
KWAN

SQL>
SQL> -- (e) Employee salary with manager salary
SQL> SELECT e.empno, e.firstname, e.lastname, e.salary,
2         m.lastname, m.salary
3 FROM OREMP e, ORDEPT d, OREMP m
4 WHERE e.workdept = REF(d)
5       AND d.mgrno = REF(m)
6 /

EMPNO  FIRSTNAME  LASTNAME      SALARY  LASTNAME      SALARY
-----
000010 CHRISTINE   HAAS          72750   HAAS          72750
000020 MICHAEL    THOMPSON      61250   THOMPSON      61250
000030 SALLY      KWAN          58250   KWAN          58250
000060 IRVING     STERN         55555   STERN         55555
000070 EVA        PULASKI       56170   STERN         55555
000050 JOHN       GEYER         60175   KWAN          58250
000090 EILEEN     HENDERSON     49750   THOMPSON      61250
000100 THEODORE   SPENSER       46150   THOMPSON      61250

8 rows selected.

```

```

SQL>
SQL> -- (f) Average salary by sex per department
SQL> SELECT d.deptno, d.deptname, e.sex, AVG(e.salary)
2 FROM OREMP e, ORDEPT d
3 WHERE e.workdept = REF(d)
4 GROUP BY d.deptno, d.deptname, e.sex
5 /

DEP DEPTNAME                                S  AVG(E.SALARY)
---
A00 SPIFFY COMPUTER SERVICE DIV.           F      72750
B01 PLANNING                               M      53700
C01 INFORMATION CENTRE                     F      58250
D01 DEVELOPMENT CENTRE                     M      55555
D01 DEVELOPMENT CENTRE                     F      56170
C01 INFORMATION CENTRE                     M      60175
B01 PLANNING                               F      49750

7 rows selected.

```

```

SQL> set termout off
SQL> set echo on
SQL> spool C:\Users\User\Downloads\prac2.out
SQL> @C:\Users\User\Downloads\script
SQL> spool off
SQL> |

```

Question 4

To drop mutually dependent types, use:

```
SQL> DROP TYPE emp_t FORCE;  
Type dropped.  
  
SQL> DROP TYPE dept_t FORCE;  
Type dropped.
```

To drop mutually dependent tables, use:

```
SQL> DROP TABLE OREMP CASCADE CONSTRAINTS;  
Table dropped.  
  
SQL> DROP TABLE ORDEPT CASCADE CONSTRAINTS;  
Table dropped.  
  
SQL>
```